

Table of Contents

1	Project Description	1
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Generating Tsunami Energy-Dissipating Barriers Using Machine Learning

1 Project Description

This studentship will extend the MECH0020 numerical framework, supervised by Dr Lama Hamadeh, by advancing from the earlier stream-function-vorticity formulation toward a benchmark tsunami model based on the nonlinear Shallow Water Equations². Inspired by tetrapods, Figure 1, and related coastal armour units used for wave-energy dissipation and coastal protection, the project will couple a validated large-scale hydrodynamic simulation to a local three-dimensional tsunami-structure interaction model in order to iteratively generate and refine barrier geometries that reduce tsunami energy transmission and structural loading³⁻⁷. The 11 March 2011 Tōhoku tsunami will serve as the primary benchmark case study, on two accounts the first being high-quality observational data are available for validation^{8,9} and second is the Sanriku/Tohoku coastline is one of the most historically tsunami-vulnerable regions in Japan, having experienced major destructive events such as the 1896 and 1933 Sanriku tsunamis, while the 2011 event itself produced extreme run-up along the same coast^{9,10,11}.



Figure 1: Concrete Tetrapod used for Coastal Erosion¹.